PHYSICAL CHEMISTRY · CHEMICAL THERMODYNAMICS

物理化学 · 化学热力学 · 习题精讲

熵、第三定律与吉布斯能

Entropy, the Third Law & Gibbs Energy

四道经典习题 · 中英双语分步讲解

Four Worked Problems · Bilingual Step-by-Step Edition

$\Delta S \geq 0$

$G = H - TS$ · $S(0) = 0$ · $dS = dq/T$

01	**理想气体等温膨胀：可逆 vs. 不可逆** *Isothermal expansion: reversible vs. irreversible*	q, w, ΔU, ΔH, ΔS, ΔS_{surr}, ΔS_{tot}
02	**从冰到过热水蒸气的熵变** *Entropy change from ice to superheated steam*	ΔS（五段可逆路径）
03	**液态汞的第三定律标准摩尔熵** *Third-Law standard molar entropy of Hg(l)*	$S_m^{\ominus}(298\ \text{K})$
04	**理想气体等温膨胀的吉布斯能变** *Gibbs energy change of isothermal expansion*	ΔG

2026–11–05

符号约定 Sign convention：Atkins（w = 外界对体系做的功 / work done *on* the system）

REFERENCE · 速查

核心公式与常数 *Key Formulas & Constants*

阅读说明 · How to read

中 每道题按「原题 → 已知 → 思路 → 分步求解 → 结果汇总 → 易错点 → 小结」展开。每段讲解先给中文，再给英文。计算过程中保留多位中间数字，最终结果按题给数据的有效数字给出。

EN Each problem runs: statement → data → strategy → step-by-step solution → results → pitfalls → takeaway. Every explanation is given in Chinese first, then in English. Intermediate values keep extra digits; final answers follow the significant figures of the data.

第一定律与功 *First Law & Work*

$$\Delta U = q + w$$

$$w_{\mathrm{rev}} = -nRT\ln\frac{V_f}{V_i}$$

等温可逆 · isothermal, reversible

$$w = -p_{\mathrm{ex}}\,\Delta V$$

恒外压 · constant external pressure

熵变 *Entropy Changes*

$$\mathrm{d}S = \frac{\mathrm{d}q_{\mathrm{rev}}}{T}$$

$$\Delta S = nR\ln\frac{V_f}{V_i} = nR\ln\frac{p_i}{p_f}$$

理想气体等温 · perfect gas, isothermal

$$\Delta S = nC_{p,m}\ln\frac{T_f}{T_i}, \qquad \Delta_{\mathrm{trs}}S = \frac{\Delta_{\mathrm{trs}}H}{T_{\mathrm{trs}}}$$

恒压升温 · heating at constant p ｜ 相变 · phase transition

环境熵与总熵 *Surroundings & Total*

$$\Delta S_{\mathrm{surr}} = -\frac{q}{T}$$

$$\Delta S_{\mathrm{tot}} = \Delta S + \Delta S_{\mathrm{surr}} \;\geq\; 0$$

$=0$：可逆 reversible ｜ >0：不可逆、自发 irreversible, spontaneous

第三定律 *Third Law*

$$S(T\to 0) = 0 \quad \text{（完美晶体 perfect crystal）}$$

$$C_{p,m} = aT^3 \;\Rightarrow\; S_m(T) = \tfrac{1}{3}\,C_{p,m}(T)$$

德拜 T^3 外推 · Debye T^3 extrapolation

吉布斯能 *Gibbs Energy*

$$G = H - TS, \qquad \left(\frac{\partial G}{\partial p}\right)_T = V$$

$$\Delta G = nRT\ln\frac{p_f}{p_i} = nRT\ln\frac{V_i}{V_f}$$

理想气体等温 · perfect gas, isothermal

常数与换算 *Constants & Conversions*

R	$8.3145\ \mathrm{J\,K^{-1}\,mol^{-1}}$
$1\ \mathrm{atm}$	$101\,325\ \mathrm{Pa}$
$1\ \mathrm{bar}$	$10^5\ \mathrm{Pa}$
T/K	$\theta/^{\circ}\mathrm{C} + 273.15$
$M(\mathrm{H_2O})$	$18.02\ \mathrm{g\,mol^{-1}}$
$1\ \mathrm{dm^3}$	$1\ \mathrm{L} = 10^{-3}\ \mathrm{m^3}$

01 理想气体等温膨胀：可逆 vs. 不可逆

Isothermal Expansion of a Perfect Gas: Reversible vs. Irreversible

第一定律 First Law　状态函数 State functions　克劳修斯不等式 Clausius inequality

原题 · Problem

A sample consisting of 1.00 mol of perfect gas molecules at 27 °C is expanded isothermally from an initial pressure of 3.00 atm to a final pressure of 1.00 atm in two ways: (a) reversibly, and (b) against a constant external pressure of 1.00 atm. Evaluate q, w, ΔU, ΔH, ΔS, ΔS_{surr}, and ΔS_{tot} in each case.

题意 · Translation

1.00 mol 理想气体在 27 °C 下等温膨胀，压强由 3.00 atm 降到 1.00 atm。膨胀按两种方式进行：(a) 可逆膨胀；(b) 对抗 1.00 atm 的恒定外压膨胀。分别求两种情况下的 q、w、ΔU、ΔH、ΔS、ΔS_{surr} 和 ΔS_{tot}。

已知 *Given*

$n = 1.00\ \text{mol}$　　$T = 27 + 273.15 = 300.15\ \text{K}$

$p_i = 3.00\ \text{atm},\ \ p_f = 1.00\ \text{atm}$　　(b) $p_{\text{ex}} = 1.00\ \text{atm}$

常用中间量 Useful number：$nRT = (1.00)(8.3145)(300.15) = 2.4956 \times 10^3\ \text{J}$

解题思路 *Strategy*

中 先分清状态函数和路径函数。ΔU、ΔH、ΔS 只取决于始态和终态。(a)(b) 两种方式的始末态完全相同，所以这三个量在两种情况下相等。q 和 w 取决于路径，必须分别计算。ΔS_{surr} 由环境实际交换的热决定，所以它也随路径变化。

EN First separate **state functions** from **path functions**. ΔU, ΔH and ΔS depend only on the initial and final states. These states are identical in (a) and (b), so the three quantities are the same in both cases. q and w depend on the path and must be calculated separately. ΔS_{surr} is set by the heat actually exchanged with the surroundings, so it also depends on the path.

共同部分 *Common to (a) and (b)*

1 ΔU 与 ΔH *ΔU and ΔH*

$$\Delta U = 0, \qquad \Delta H = \Delta U + \Delta(pV) = 0 + \Delta(nRT) = 0$$

中 理想气体分子之间没有相互作用，内能只是温度的函数。等温过程中 T 不变，所以 $\Delta U = 0$。同时 $pV = nRT$ 也不变，所以 $\Delta H = 0$。

EN Perfect-gas molecules do not interact, so the internal energy depends only on temperature. T is constant here, so $\Delta U = 0$. Since $pV = nRT$ is also constant, $\Delta H = 0$.

2 体系熵变 ΔS *Entropy change of the system*

$$\Delta S = nR\ln\frac{V_f}{V_i} = nR\ln\frac{p_i}{p_f} = (1.00\ \text{mol})(8.3145\ \text{J K}^{-1}\,\text{mol}^{-1})\ln 3 = \boxed{+9.13\ \text{J K}^{-1}}$$

中 熵是状态函数。即使 (b) 是不可逆过程，也可以在相同的始末态之间设计一条可逆等温路径来计算 ΔS，所以这个结果对 (a)(b) 都成立。等温时由玻意耳定律，$V_f/V_i = p_i/p_f = 3$。

EN Entropy is a state function. Even though (b) is irreversible, ΔS can be calculated along a reversible isothermal path between the same two states, so this result holds for both (a) and (b). At constant temperature Boyle's law gives $V_f/V_i = p_i/p_f = 3$.

(a) 可逆膨胀 *Reversible expansion*

3 功与热 *Work and heat*

$$w = -\int_{V_i}^{V_f} p\,\mathrm{d}V = -nRT\ln\frac{V_f}{V_i} = -(2.4956\times10^{3}\ \text{J})\ln 3 = \boxed{-2.74\ \text{kJ}}$$

$$q = \Delta U - w = 0 - (-2.74\ \text{kJ}) = \boxed{+2.74\ \text{kJ}}$$

中 可逆膨胀时，外压始终只比气体压强小一个无穷小量，可以取 $p_{\text{ex}} = p = nRT/V$ 积分。$w<0$ 表示气体对外做功。因为 $\Delta U = 0$，这部分能量全部来自从环境吸收的热，所以 $q = -w$。

EN In a reversible expansion the external pressure is always only infinitesimally below the gas pressure, so we can set $p_{\text{ex}} = p = nRT/V$ and integrate. $w<0$ means the gas does work on the surroundings. Because $\Delta U = 0$, all of that energy comes in as heat from the surroundings, so $q = -w$.

4 环境熵变与总熵变 *Surroundings and total*

$$\Delta S_{\text{surr}} = -\frac{q}{T} = -\frac{2.74\times10^{3}\ \text{J}}{300.15\ \text{K}} = \boxed{-9.13\ \text{J K}^{-1}},\qquad \Delta S_{\text{tot}} = 9.13 - 9.13 = \boxed{0}$$

中 环境放出的热正好等于体系吸收的热 q，所以环境的熵减少 q/T。可逆过程的总熵变为零，这就是热力学第二定律意义上的"可逆"。

EN The surroundings give up exactly the heat q that the system absorbs, so their entropy falls by q/T. The total entropy change of a reversible process is zero, which is what "reversible" means in Second-Law terms.

(b) 恒外压膨胀 *Expansion against constant* p_{ex} = 1.00 atm

5 功与热 *Work and heat*

$$V_i = \frac{nRT}{p_i} = \frac{2.4956\times10^{3}\ \text{J}}{3.00\times101\,325\ \text{Pa}} = 8.21\ \text{dm}^3,\qquad V_f = \frac{nRT}{p_f} = 24.63\ \text{dm}^3$$

$$w = -p_{\text{ex}}\Delta V = -(101\,325\ \text{Pa})(16.42\times10^{-3}\ \text{m}^3) = \boxed{-1.66\ \text{kJ}},\qquad q = -w = \boxed{+1.66\ \text{kJ}}$$

简便写法 Shortcut：$w = -p_{\text{ex}}\,nRT\left(\frac{1}{p_f}-\frac{1}{p_i}\right) = -\left(1-\frac{1}{3}\right)nRT = -\frac{2}{3}nRT = -1.66\ \text{kJ}$

中 外压恒定时，功等于外压乘以体积变化，与气体内部压强如何变化无关。因为 $p_{\text{ex}} = p_f$，结果可以化简为 $-\frac{2}{3}nRT$。它比可逆功小：同样的始末态，可逆膨胀对外做的功最多。

EN With a constant external pressure, the work is the external pressure times the volume change, whatever happens to the pressure inside the gas. Because $p_{\text{ex}} = p_f$, the result simplifies to $-\frac{2}{3}nRT$. It is smaller in magnitude than the reversible work: between the same two states, a reversible expansion delivers the most work.

6 环境熵变与总熵变 *Surroundings and total*

$$\Delta S_{\text{surr}} = -\frac{q}{T} = -\frac{1.66\times10^{3}\ \text{J}}{300.15\ \text{K}} = \boxed{-5.54\ \text{J K}^{-1}},\qquad \Delta S_{\text{tot}} = 9.13 - 5.54 = \boxed{+3.59\ \text{J K}^{-1}} > 0$$

中 这里 $\Delta S_{\text{surr}} \neq -\Delta S$。环境相当于一个大热源，它的熵变只取决于它实际传出的热 $q = 1.66\ \text{kJ}$。$\Delta S_{\text{tot}} > 0$ 说明这是自发的不可逆过程，符合克劳修斯不等式。

EN Here $\Delta S_{\text{surr}} \neq -\Delta S$. The surroundings behave as a large heat reservoir, so their entropy change depends only on the heat they actually transfer, $q = 1.66\ \text{kJ}$. $\Delta S_{\text{tot}} > 0$ shows the process is spontaneous and irreversible, as the Clausius inequality requires.

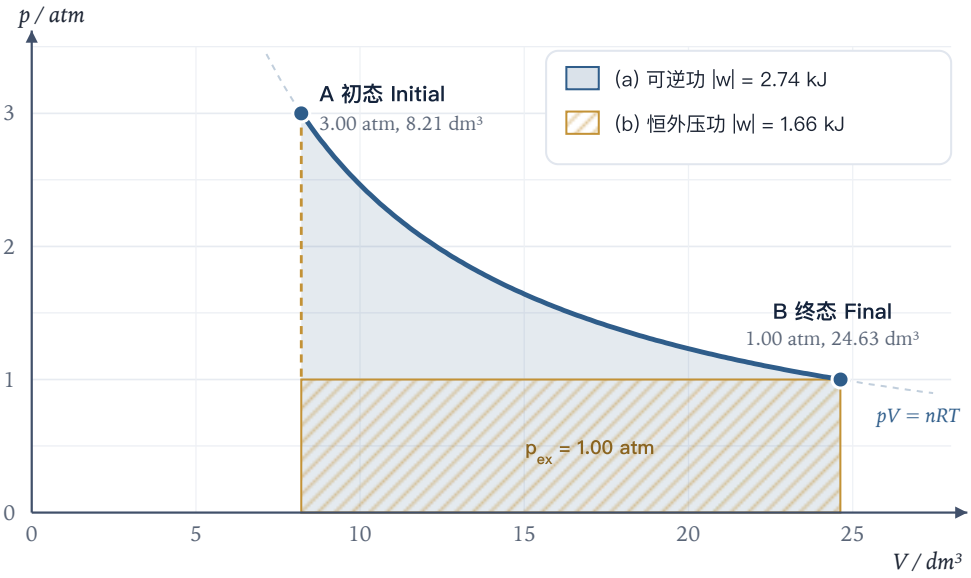


图 1　p–V 图：等温线下的面积是可逆功 $|w_{(a)}| = 2.74\ \text{kJ}$，阴影矩形是恒外压功 $|w_{(b)}| = 1.66\ \text{kJ}$。

Figure 1. The area under the isotherm is the reversible work, $|w_{(a)}| = 2.74\ \text{kJ}$; the hatched rectangle is the work against constant p_{ex}, $|w_{(b)}| = 1.66\ \text{kJ}$.

结果汇总 *Results*

物理量 Quantity	(a) 可逆 Reversible	(b) 恒外压 Constant p_{ex}	性质 Type
q	$+2.74\ \text{kJ}$	$+1.66\ \text{kJ}$	路径 path
w	$-2.74\ \text{kJ}$	$-1.66\ \text{kJ}$	路径 path
ΔU	0	0	状态 state
ΔH	0	0	状态 state
ΔS	$+9.13\ \text{J K}^{-1}$	$+9.13\ \text{J K}^{-1}$	状态 state
ΔS_{surr}	$-9.13\ \text{J K}^{-1}$	$-5.54\ \text{J K}^{-1}$	随 q 而变 follows q
ΔS_{tot}	0	$+3.59\ \text{J K}^{-1}$	判据 criterion

! 易错点 · Pitfalls

1. 中 (b) 的 ΔS 不能用 q/T 计算（$1.66\ \mathrm{kJ}/300.15\ \mathrm{K} = 5.54\ \mathrm{J\,K^{-1}}$ 是错的）。$\mathrm{d}S = \mathrm{d}q/T$ 只对可逆热成立。

   EN In (b), $\Delta S \neq q/T$; the value $5.54\ \mathrm{J\,K^{-1}}$ is wrong for the system. $\mathrm{d}S = \mathrm{d}q/T$ holds only for reversible heat.

2. 中 温度必须用开尔文：$27\ ^\circ\mathrm{C} = 300.15\ \mathrm{K}$。

   EN Use kelvin: $27\ ^\circ\mathrm{C} = 300.15\ \mathrm{K}$.

3. 中 注意 w 的符号：本讲义中 w 是外界对体系做的功，膨胀时 $w < 0$。

   EN Mind the sign of w: here w is the work done on the system, so $w < 0$ for an expansion.

✓ 一句话总结 · Takeaway

中 **状态函数只看始末态，路径函数要看过程；不可逆性的"代价"体现在 $\Delta S_{\mathrm{tot}} > 0$。**

EN State functions depend only on the end states, path functions depend on the process, and the cost of irreversibility appears as $\Delta S_{\mathrm{tot}} > 0$.

02 从冰到过热水蒸气的熵变

Entropy Change from Ice to Superheated Steam

变温熵变 Heating　相变熵 Phase transitions　设计可逆路径 Reversible path

原题 · Problem

Calculate the change in entropy of the system when 10.0 g of ice at −10.0 °C is converted into water vapour at 115.0 °C and at a constant pressure of 1 bar. The molar constant-pressure heat capacities are: $C_{p,\mathrm{m}}(\mathrm{H_2O(s)}) = 37.6\ \mathrm{J\ K^{-1}\ mol^{-1}}$; $C_{p,\mathrm{m}}(\mathrm{H_2O(l)}) = 75.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$; and $C_{p,\mathrm{m}}(\mathrm{H_2O(g)}) = 33.6\ \mathrm{J\ K^{-1}\ mol^{-1}}$. The standard enthalpy of vaporization of $\mathrm{H_2O(l)}$ is 40.7 kJ mol^{-1}, and the standard enthalpy of fusion of $\mathrm{H_2O(l)}$ is 6.01 kJ mol^{-1}, both at the relevant transition temperatures.

题意 · Translation

10.0 g、−10.0 °C 的冰在 1 bar 恒压下转变为 115.0 °C 的水蒸气，求体系的熵变。已知摩尔定压热容：冰 37.6、液态水 75.3、水蒸气 33.6（单位均为 $\mathrm{J\,K^{-1}\,mol^{-1}}$）。$H_2O$ 的标准摩尔蒸发焓为 40.7 kJ mol^{-1}，标准摩尔熔化焓为 6.01 kJ mol^{-1}，两者均为对应相变温度下的数值。

已知 *Given*

$$n = \frac{10.0\ \mathrm{g}}{18.02\ \mathrm{g\,mol^{-1}}} = 0.5549\ \mathrm{mol}$$

$$T_i = 263.15\ \mathrm{K},\quad T_f = 388.15\ \mathrm{K}$$

$$T_{\mathrm{fus}} = 273.15\ \mathrm{K},\quad T_{\mathrm{b}} = 373.15\ \mathrm{K}$$

$$\Delta_{\mathrm{fus}}H_m = 6010\ \mathrm{J\,mol^{-1}},\quad \Delta_{\mathrm{vap}}H_m = 40\,700\ \mathrm{J\,mol^{-1}}$$

解题思路 *Strategy*

中 熵是状态函数，所以可以用五个可逆步骤代替实际过程：冰升温 → 熔化 → 水升温 → 汽化 → 蒸汽升温。升温段用 $\int C_p\,\mathrm{d}T/T$，相变段用 $\Delta H/T$，最后把五段相加。

EN Because entropy is a state function, the real process can be replaced by five reversible steps: warm the ice → melt it → warm the water → vaporize it → warm the vapour. Use $\int C_p\,\mathrm{d}T/T$ for each heating stage and $\Delta H/T$ for each phase transition, then add the five terms.

冰 ice −10 °C　① 升温 →　冰 ice 0 °C　② 熔化 →　水 water 0 °C　③ 升温 →　水 water 100 °C　④ 汽化 →　汽 steam 100 °C　⑤ 升温 →　汽 steam 115 °C

0 两个基本公式 *The two building blocks*

$$\Delta S_{\mathrm{heat}} = \int_{T_1}^{T_2} \frac{nC_{p,m}\,\mathrm{d}T}{T} = nC_{p,m}\ln\frac{T_2}{T_1},\qquad \Delta S_{\mathrm{trs}} = \frac{n\,\Delta_{\mathrm{trs}}H_m}{T_{\mathrm{trs}}}$$

中 恒压下 $\mathrm{d}q_{\mathrm{rev}} = nC_{p,m}\mathrm{d}T$，热容视为常数时积分得到对数形式。相变在相变温度和恒压下可逆进行，吸收的热等于 ΔH，所以 $\Delta S = \Delta H/T$。

EN At constant pressure $\mathrm{d}q_{\mathrm{rev}} = nC_{p,m}\mathrm{d}T$; with a constant heat capacity the integral gives the logarithm. A phase transition at its transition temperature and constant pressure is reversible and absorbs heat equal to ΔH, so $\Delta S = \Delta H/T$.

1 冰升温 263.15 → 273.15 K　*Warm the ice*

$$\Delta S_1 = (0.5549)(37.6)\ln\frac{273.15}{263.15} = +0.778\ \mathrm{J\,K^{-1}}$$

2 熔化 273.15 K　*Fusion*

$$\Delta S_2 = \frac{(0.5549)(6010)}{273.15} = +12.21\ \mathrm{J\,K^{-1}}$$

3 水升温 273.15 → 373.15 K　*Warm the liquid*

$$\Delta S_3 = (0.5549)(75.3)\ln\frac{373.15}{273.15} = +13.04\ \mathrm{J\,K^{-1}}$$

4 汽化 373.15 K　*Vaporization*

$$\Delta S_4 = \frac{(0.5549)(40\,700)}{373.15} = +60.53\ \mathrm{J\,K^{-1}}$$

5 蒸汽升温 373.15 → 388.15 K　*Warm the vapour*

$$\Delta S_5 = (0.5549)(33.6)\ln\frac{388.15}{373.15} = +0.735\ \mathrm{J\,K^{-1}}$$

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 + \Delta S_5 = 0.778 + 12.21 + 13.04 + 60.53 + 0.735 = \boxed{+87.3\ \mathrm{J\,K^{-1}}}$$

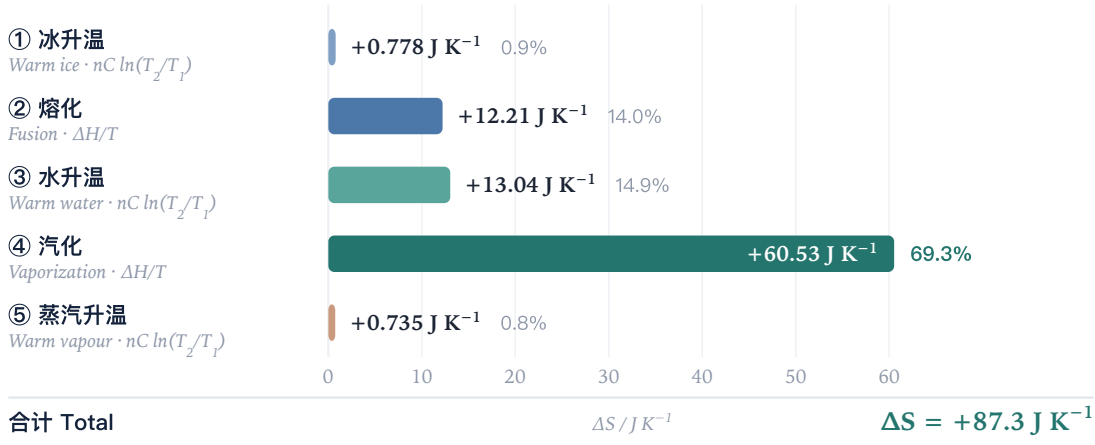


图 2　五个步骤对总熵变的贡献。汽化一步约占 69%。

Figure 2. Contribution of each step to the total entropy change. Vaporization alone accounts for about 69%.

i 物理图像 · Physical insight

中　液体变成气体时，分子可以占据的空间急剧增大，无序度增加最多，所以汽化熵远大于熔化熵。水的摩尔蒸发熵 $40\,700/373.15 = 109\ \mathrm{J\,K^{-1}\,mol^{-1}}$，高于楚顿规则的约 $85\ \mathrm{J\,K^{-1}\,mol^{-1}}$：液态水中的氢键使液相更有序，汽化时熵增就更大。

EN　When a liquid turns into a gas, the volume available to the molecules increases enormously, so the gain in disorder is largest; this is why the entropy of vaporization far exceeds that of fusion. Water's molar entropy of vaporization, $40\,700/373.15 = 109\ \mathrm{J\,K^{-1}\,mol^{-1}}$, is higher than Trouton's value of about $85\ \mathrm{J\,K^{-1}\,mol^{-1}}$: hydrogen bonding makes liquid water more ordered, so vaporizing it produces a larger entropy gain.

! 易错点 · Pitfalls

1. 中 不能把总焓变除以某一个温度，因为热是在一段温度范围内传递的。

   EN Do not divide the total enthalpy change by a single temperature; the heat is transferred over a range of temperatures.

2. 中 焓的单位是 kJ mol^{-1}，代入前换成 J mol^{-1}（6.01 kJ = 6010 J），否则结果差 1000 倍。

   EN Convert kJ mol^{-1} to J mol^{-1} before dividing (6.01 kJ = 6010 J), or the answer is off by a factor of 1000.

3. 中 每一段要用当时那一相的热容：冰 37.6、水 75.3、蒸汽 33.6。

   EN In each stage use the heat capacity of the phase actually present: ice 37.6, water 75.3, vapour 33.6.

4. 中 题目给的是质量，先换算成物质的量 n。

   EN The data give a mass; convert it to the amount of substance n first.

✓ 一句话总结 · Takeaway

中 沿可逆路径分段累加熵变：升温用 $nC_{p,m}\ln(T_2/T_1)$，相变用 $\Delta H/T$。

EN Build ΔS piece by piece along a reversible path: $nC_{p,m}\ln(T_2/T_1)$ for heating, $\Delta H/T$ for transitions.

03 液态汞的第三定律标准摩尔熵

Third-Law Standard Molar Entropy of Liquid Mercury

热力学第三定律 Third Law　德拜外推 Debye extrapolation　绝对熵 Absolute entropy

原题 · Problem

At 10 K $C_{p,\mathrm{m}}(\mathrm{Hg(s)}) = 4.64\ \mathrm{J\ K^{-1}\ mol^{-1}}$. Between 10 K and the melting point of Hg(s), 234.3 K, heat capacity measurements indicate that the entropy increases by $57.74\ \mathrm{J\ K^{-1}\ mol^{-1}}$. The standard enthalpy of fusion of Hg(s) is $2322\ \mathrm{J\ mol^{-1}}$ at 234.3 K. Between the melting point and 298.0 K, heat capacity measurements indicate that the entropy increases by $6.85\ \mathrm{J\ K^{-1}\ mol^{-1}}$. Determine the Third-Law standard molar entropy of Hg(l) at 298 K.

题意 · Translation

10 K 时固态汞的 $C_{p,m} = 4.64\,\mathrm{J\,K^{-1}\,mol^{-1}}$。热容测量表明，从 10 K 到汞的熔点 234.3 K，熵增加 $57.74\,\mathrm{J\,K^{-1}\,mol^{-1}}$；234.3 K 时汞的标准熔化焓为 $2322\ \mathrm{J\ mol^{-1}}$；从熔点到 298.0 K，熵又增加 $6.85\,\mathrm{J\,K^{-1}\,mol^{-1}}$。求 298 K 时 Hg(l) 的第三定律标准摩尔熵。

解题思路 *Strategy*

中 第三定律规定完美晶体在 0 K 时熵为零，因此 $S_m(298\ \mathrm{K})$ 就是从 0 K 到 298 K 所有熵变之和。实验热容只测到 10 K，0–10 K 这一段用德拜 T^3 定律外推。

EN The Third Law sets the entropy of a perfect crystal to zero at 0 K, so $S_m(298\ \mathrm{K})$ equals the sum of all entropy changes from 0 K to 298 K. Heat-capacity data reach down only to 10 K, so the 0–10 K segment is obtained from the Debye T^3 law.

$$S_m^{\ominus}(298\ \mathrm{K}) = \underbrace{S_m(10\ \mathrm{K})}_{\text{德拜 Debye}} + \underbrace{\Delta S(10 \to 234.3\ \mathrm{K})}_{\text{固体升温 solid}} + \underbrace{\frac{\Delta_{\mathrm{fus}}H^{\ominus}}{T_{\mathrm{fus}}}}_{\text{熔化 fusion}} + \underbrace{\Delta S(234.3 \to 298\ \mathrm{K})}_{\text{液体升温 liquid}}$$

1 0 → 10 K：德拜外推 *Debye extrapolation*

$$C_{p,m} \approx aT^3 \ \Rightarrow\ S_m(T) = \int_0^T \frac{aT'^3}{T'}\,\mathrm{d}T' = \frac{aT^3}{3} = \frac{C_{p,m}(T)}{3}$$

$$S_m(10\ \mathrm{K}) = \frac{4.64}{3} = 1.55\ \mathrm{J\,K^{-1}\,mol^{-1}}$$

中 极低温下固体的热容近似遵循 $C_p \propto T^3$（德拜 T^3 定律）。代入 $\int C_p\,\mathrm{d}T/T$，被积函数是 aT^2，积分出 $1/3$ 因子，得到 $S = C_p/3$，不需要单独求系数 a。

EN At very low temperatures the heat capacity of a solid follows $C_p \propto T^3$ (the Debye T^3 law). In $\int C_p\,\mathrm{d}T/T$ the integrand becomes aT^2, which integrates to give a factor of $1/3$, so $S = C_p/3$ and the coefficient a is never needed.

注：严格地说，金属在极低温下还有一项与 T 成正比的电子热容；本题按教材惯例只用德拜项。

Note: strictly, a metal also has an electronic heat capacity proportional to T at very low temperatures; following the textbook convention, only the Debye term is used here.

2 10 → 234.3 K：固体升温 *Heating the solid*

$$\Delta S = \int_{10\ \mathrm{K}}^{234.3\ \mathrm{K}} \frac{C_{p,m}}{T}\,\mathrm{d}T = +57.74\ \mathrm{J\,K^{-1}\,mol^{-1}} \qquad (\text{题给 given})$$

3　234.3 K：熔化 *Fusion*

$$\Delta_{\mathrm{fus}}S^{\ominus} = \frac{\Delta_{\mathrm{fus}}H^{\ominus}}{T_{\mathrm{fus}}} = \frac{2322\ \mathrm{J\,mol^{-1}}}{234.3\ \mathrm{K}} = +9.91\ \mathrm{J\,K^{-1}\,mol^{-1}}$$

4　234.3 → 298 K：液体升温 *Heating the liquid*

$$\Delta S = +6.85\ \mathrm{J\,K^{-1}\,mol^{-1}} \quad （题给 given）$$

$$S_m^{\ominus}(\mathrm{Hg,\,l,\,298\ K}) = 1.55 + 57.74 + 9.91 + 6.85 = 76.05 \approx \boxed{76.0\ \mathrm{J\,K^{-1}\,mol^{-1}}}$$

中　这个结果与数据表中 Hg(l) 的标准摩尔熵（约 $76\ \mathrm{J\,K^{-1}\,mol^{-1}}$）吻合。

EN　This agrees with the tabulated standard molar entropy of Hg(l), about $76\ \mathrm{J\,K^{-1}\,mol^{-1}}$.

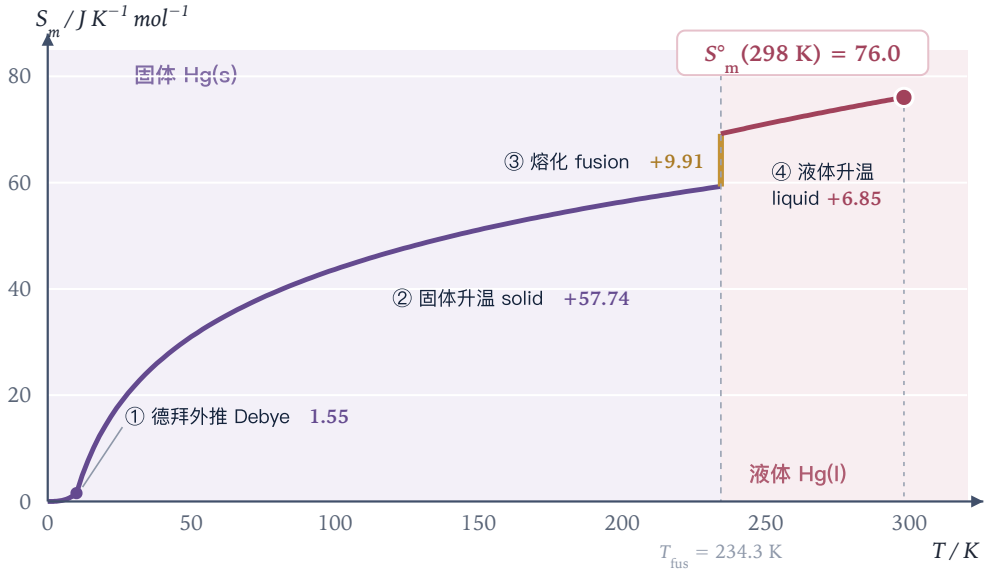


图 3　Hg 的摩尔熵随温度的累积：熔点处的竖直跳跃就是熔化熵。两数据点之间的曲线形状为示意。

Figure 3. Cumulative molar entropy of Hg versus temperature; the vertical jump at the melting point is the entropy of fusion. The curve shape between the given values is schematic.

! 易错点 · Pitfalls

1. 中　不要漏掉 0–10 K 这一段。它只有 1.55，但第三定律熵必须从 0 K 算起。
   EN　Do not skip the 0–10 K segment. It contributes only 1.55, but a Third-Law entropy must start from 0 K.
2. 中　德拜外推给出的是 $C_p/3$，不是 C_p。
   EN　The Debye extrapolation gives $C_p/3$, not C_p.
3. 中　第三定律熵是以 $S(0)=0$ 为零点的绝对值，它总是正的；不要和过程的 ΔS 混淆。
   EN　A Third-Law entropy is an absolute value referenced to $S(0)=0$ and is always positive; do not confuse it with the ΔS of a process.

✓ 一句话总结 · Takeaway

中　$S_m^{\ominus}$ = 德拜外推 + $\int C_p\,\mathrm{d}T/T$ + 各相变的 $\Delta_{\mathrm{trs}}H/T_{\mathrm{trs}}$。

EN　$S_m^{\ominus}$ = Debye extrapolation + $\int C_p\,\mathrm{d}T/T + \Delta_{\mathrm{trs}}H/T_{\mathrm{trs}}$ for every transition.

04 理想气体等温膨胀的吉布斯能变

Gibbs Energy Change of an Isothermal Expansion

吉布斯能 Gibbs energy　$(\partial G/\partial p)_T = V$　交叉验证 Cross-check

原题 · Problem

Suppose that 2.5 mmol of perfect gas molecules initially occupies 42 cm^3 at 300 K and then expands isothermally to 600 cm^3. Calculate ΔG for the process.

题意 · Translation

2.5 mmol 理想气体在 300 K 时最初占据 42 cm^3，随后等温膨胀到 600 cm^3。求该过程的 ΔG。

已知 *Given*

$n = 2.5\ \text{mmol} = 2.5 \times 10^{-3}\ \text{mol}$　　$T = 300\ \text{K}$（等温 isothermal）

$V_i = 42\ \text{cm}^3$　　$V_f = 600\ \text{cm}^3$

解题思路 *Strategy*

中 恒温时，G 随压强的变化由 $(\partial G/\partial p)_T = V$ 给出。把理想气体的 $V = nRT/p$ 代入积分，再用玻意耳定律把压强比换成体积比。

EN At constant temperature, the pressure dependence of G is given by $(\partial G/\partial p)_T = V$. Substitute $V = nRT/p$ for a perfect gas, integrate, and use Boyle's law to turn the pressure ratio into a volume ratio.

1 推导公式 *Derive the expression*

$$\mathrm{d}G = V\,\mathrm{d}p - S\,\mathrm{d}T \xrightarrow{\mathrm{d}T=0} \Delta G = \int_{p_i}^{p_f} \frac{nRT}{p}\,\mathrm{d}p = nRT\ln\frac{p_f}{p_i} = nRT\ln\frac{V_i}{V_f}$$

中 最后一步用了玻意耳定律 $p_iV_i = p_fV_f$，即 $p_f/p_i = V_i/V_f$。

EN The last step uses Boyle's law, $p_iV_i = p_fV_f$, so $p_f/p_i = V_i/V_f$.

2 代入数值 *Substitute the data*

$$\Delta G = (2.5 \times 10^{-3}\ \text{mol})(8.3145\ \text{J K}^{-1}\,\text{mol}^{-1})(300\ \text{K})\ln\frac{42}{600}$$

$$= (6.236\ \text{J}) \times (-2.659) = -16.6\ \text{J} \approx \boxed{-17\ \text{J}}$$

中 体积比没有量纲，cm^3 不必换算成 m^3。题给的 2.5 mmol 和 42 cm^3 都只有两位有效数字，所以结果写作 -17 J。

EN The volume ratio is dimensionless, so there is no need to convert cm^3 to m^3. The data 2.5 mmol and 42 cm^3 have two significant figures, so the answer is reported as -17 J.

3 交叉验证：$G = H - TS$ *Cross-check*

方法一 Method 1：$\int V\,\mathrm{d}p$

$$\Delta G = nRT\ln\frac{V_i}{V_f} = -16.6\ \mathrm{J}$$

方法二 Method 2：$\Delta H - T\Delta S$

$$\Delta H = 0,\quad \Delta S = nR\ln\frac{V_f}{V_i} = +5.53\times10^{-2}\ \mathrm{J\,K^{-1}}$$

$$\Delta G = 0 - (300)(5.53\times10^{-2}) = -16.6\ \mathrm{J}\ \checkmark$$

中 两种方法结果一致。理想气体等温过程的 $\Delta H = 0$，所以 ΔG 完全来自熵项 $-T\Delta S$：气体膨胀使熵增加，吉布斯能随之降低。

EN Both methods agree. For an isothermal perfect gas $\Delta H = 0$, so ΔG comes entirely from the entropy term $-T\Delta S$: the expansion raises the entropy and lowers the Gibbs energy accordingly.

! 易错点 · Pitfalls

1. 中 对数里是 V_i/V_f，不是 V_f/V_i。膨胀使压强降低，G 减小，ΔG 应为负值，可以用这一点检查符号。

   EN The ratio is V_i/V_f, not V_f/V_i. Expansion lowers the pressure and therefore G, so ΔG must be negative; use this to check the sign.

2. 中 "$\Delta G < 0$ 即自发"这一判据只适用于恒温、恒压且无非体积功的过程。本题压强在变化，这里的 ΔG 只是状态函数的差值，不能直接当作自发性判据。

   EN The criterion "$\Delta G < 0$ means spontaneous" applies only at constant temperature and pressure with no non-expansion work. Here the pressure changes, so ΔG is just a state-function difference, not a spontaneity criterion.

3. 中 单位换算：$2.5\ \mathrm{mmol} = 2.5\times10^{-3}\ \mathrm{mol}$。

   EN Unit conversion: $2.5\ \mathrm{mmol} = 2.5\times10^{-3}\ \mathrm{mol}$.

✓ 一句话总结 · Takeaway

中 理想气体等温变化：$\Delta G = nRT\ln(p_f/p_i) = -T\Delta S$。

EN For an isothermal perfect gas: $\Delta G = nRT\ln(p_f/p_i) = -T\Delta S$.

SUMMARY · 总结

答案总览 *Answer Summary*

题号 No.	所求 Quantity	答案 Answer	关键方法 Key method
1	(a) 可逆 reversible	$q=+2.74\ \mathrm{kJ},\ w=-2.74\ \mathrm{kJ}$ $\Delta U=\Delta H=0,\ \Delta S=+9.13\ \mathrm{J\,K^{-1}}$ $\Delta S_{\mathrm{surr}}=-9.13\ \mathrm{J\,K^{-1}},\ \Delta S_{\mathrm{tot}}=0$	状态函数与路径函数分开处理；$\Delta S_{\mathrm{surr}}=-q/T$ *Treat state and path functions separately*
	(b) 恒外压 constant p_{ex}	$q=+1.66\ \mathrm{kJ},\ w=-1.66\ \mathrm{kJ}$ $\Delta U=\Delta H=0,\ \Delta S=+9.13\ \mathrm{J\,K^{-1}}$ $\Delta S_{\mathrm{surr}}=-5.54\ \mathrm{J\,K^{-1}},\ \Delta S_{\mathrm{tot}}=+3.59\ \mathrm{J\,K^{-1}}$	
2	ΔS（冰 → 蒸汽）	$\boxed{+87.3\ \mathrm{J\,K^{-1}}}$	五段可逆路径累加 *Sum over a five-step reversible path*
3	$S_m^{\ominus}(\mathrm{Hg,l,298\ K})$	$\boxed{76.0\ \mathrm{J\,K^{-1}\,mol^{-1}}}$	德拜外推 + 实测熵增 + 熔化熵 *Debye + measured ΔS + fusion*
4	ΔG	$\boxed{-17\ \mathrm{J}}$（$-16.6\ \mathrm{J}$）	$\Delta G=nRT\ln(V_i/V_f)$ *Integrate* $(\partial G/\partial p)_T=V$

三条主线 *Three ideas behind all four problems*

I

中 **熵是状态函数。**实际过程不可逆时，就在相同始末态之间设计一条可逆路径来计算 ΔS（第 1、2 题）。

EN **Entropy is a state function.** When the real process is irreversible, calculate ΔS along a reversible path between the same end states (Problems 1 and 2).

II

中 **环境熵看实际交换的热。**$\Delta S_{\mathrm{surr}}=-q_{\mathrm{actual}}/T$，再用 $\Delta S_{\mathrm{tot}}\geq 0$ 判断可逆性（第 1 题）。

EN **The surroundings respond to the actual heat.** $\Delta S_{\mathrm{surr}}=-q_{\mathrm{actual}}/T$; then $\Delta S_{\mathrm{tot}}\geq 0$ decides reversibility (Problem 1).

III

中 **第三定律给出绝对熵，吉布斯能把焓和熵联系起来。**$S(0)=0$ 让熵可以取绝对值；$G=H-TS$ 让我们用两种方法相互验证（第 3、4 题）。

EN **The Third Law gives absolute entropies, and the Gibbs energy links enthalpy and entropy.** $S(0)=0$ makes absolute entropies possible; $G=H-TS$ lets two methods check each other (Problems 3 and 4).

计算中保留中间位数，最终结果按题给数据的有效数字给出。所用常数：$R=8.3145\ \mathrm{J\,K^{-1}\,mol^{-1}}$，$1\ \mathrm{atm}=101\,325\ \mathrm{Pa}$，$M(\mathrm{H_2O})=18.02\ \mathrm{g\,mol^{-1}}$。
Intermediate values carry extra digits; final answers follow the significant figures of the data.